

Homework 3

Math 742

2026-02-10

Casella/Berger - Problems

7.1 One observation is taken on a discrete random variable X with pmf $f(x|\theta)$, where $\theta \in 1, 2, 3$. Find the MLE of θ .

x	$f(x 1)$	$f(x 2)$	$f(x 3)$
0	1/3	1/4	0
1	1/3	1/4	0
2	0	1/4	1/4
3	1/6	1/4	1/2
4	1/6	0	1/4

7.11 Let $X_1, X_2, \dots, X_n$ be iid with pdf

$$f(x|\theta) = \theta x^{\theta-1}, 0 \leq x \leq 1, 0 < \theta < \infty$$

- (a) Find the MLE of θ , and show that its variance $\rightarrow 0$ as $n \rightarrow \infty$
- (b) Find the method of moments estimator of θ .

Wasserman Section 9.14 - Problems

2. Let $X_1, X_2, \dots, X_n \sim \text{Uniform}(a, b)$, where a and b are unknown parameters and $a < b$.

- (a) Find the method of moments estimators for a and b .
- (b) Find the MLE estimates of a and b .
- (c) Let $\tau = \int x dF(x)$. Find the MLE of τ .
- (d) Let $\hat{\tau}$ be the MLE of τ . Let $\tilde{\tau}$ be the non-parametric plug-in estimator of $\tau = \int x dF(x)$. Suppose that $a = 1, b = 3$, and $n = 10$. Find the MSE of $\hat{\tau}$ by simulation. Find the MSER of $\tilde{\tau}$ analytically and compare to the simulated value.

6a. Let $X_1, X_2, \dots, X_n \sim N(\theta, 1)$. Define,

$$Y_i = \begin{cases} 1 & \text{if } X_i > 0 \\ 0 & \text{if } X_i \leq 0 \end{cases}$$

Let $\psi = \mathbb{P}(Y_1 = 1)$

- (a) Find the MLE for ψ .

Problem 5. Determine the MoM for the Weibull distribution whose PDF is:

Shape $k > 0$ and scale $\lambda > 0$ is

$$f(x | k, \lambda) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{x}{\lambda}\right)^k\right], & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Problem 6. MLE and Empirical MSE for the Normal Distribution

Let $X_1, \dots, X_n$ be iid from a normal distribution with unknown mean μ and **known** variance σ^2 .

1. Write the likelihood function $L(\mu)$ and log-likelihood $\ell(\mu)$.
2. Show that maximizing $\ell(\mu)$ is equivalent to **minimizing the empirical mean squared error**

$$\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

3. Use this result to find the MLE $\hat{\mu}$.